

Alina C. Spera

ACADEMIC APPOINTMENTS

NRC Postdoctoral Associate,

U.S. Naval Research Laboratory – Remote Sensing Division, 2026-Present
National Academies of Sciences, Engineering, and Medicine Fellowship

Postdoctoral Investigator,

Woods Hole Oceanographic Institution – Applied Ocean Physics and Engineering, 2024-2026
Project: *Collaborative Research: Drivers and Biogeochemical Implications of Saltwater Intrusion Along Arctic Coastlines (NSF Arctic Natural Sciences)*

EDUCATION

Ph.D. Ecology and Evolutionary Biology, 2024

University of Texas at El Paso

Advisor: Vanessa L. Lougheed. Defended July 2024

Dissertation: [Ecological, Climatic, and Geomorphic Controls of the Flux and Fate of CO₂ From Coastal Arctic Ecosystems](#)

M.S. Oceanography and Coastal Sciences, 2019

Louisiana State University

Advisor: John R. White. Defended May 2019

Thesis: [Decadal changes of soil physiochemical properties in a freshwater wetland after hydrologic reconnection](#)

B.S. Marine Science and Biology, Minor: Geological Sciences, 2016

University of Miami, Rosenstiel School of Marine and Atmospheric Science
Departmental Honors

Guest Student, Spring 2015

James Cook University, Townsville, QLD, Australia

PUBLICATIONS (peer-reviewed)

6. Marvy, A., Andresen, C., **Spera, A.C.**, Lougheed, V.L., 2026. Contrasting CO₂ fluxes among Arctic pond types are driven by differences in carbon lability, terrestrial inputs, and benthic respiration. JGR Biogeosciences. <https://doi.org/10.1029/2025JG009318>
5. **Spera, A.C.**, Peterson, S., Tweedie, C., Lougheed, V., 2026. Wind-driven Circulation Impacts Water Quality and pCO₂ Over an Estuarine Gradient in an Arctic Lagoon. Estuaries and Coasts 49. <https://doi.org/10.1007/s12237-025-01662-6>
4. **Spera, A.C.**, McClelland, J.W., Demir, C., Cardenas, M.B., Guimond, J.A., 2025. Saltwater inundation's concurrent effects on below-ground thermal hydrology and above-ground properties of Arctic coastal tundra. Science of The Total Environment 1008, 180906. <https://doi.org/10.1016/j.scitotenv.2025.180906>
3. **Spera, A.C.**, Lougheed, V.L., 2025. Carbon dioxide fluxes of Arctic coastal ecosystems controlled by seasonal patterns of land-to-ocean connectivity. Limnology and Oceanography, June 2025. <https://doi.org/10.1002/lno.70101>

2. **Spera, A.C.**, Lougheed, V.L., 2025. CO₂ Emissions From Low Order Tundra Streams Stimulated by Surface-Subsurface Connectivity Following Extreme Rainfall Events. JGR Biogeosciences 130, e2024JG008459. <https://doi.org/10.1029/2024JG008459>
1. **Spera, A.C.**, White, J.R., Corstanje, R., 2020. Spatial and temporal changes to a hydrologically-reconnected coastal wetland: Implications for restoration. Estuarine, Coastal and Shelf Science 238, 106728. <https://doi.org/10.1016/j.ecss.2020.106728>

GRANTS + FELLOWSHIPS

- **Co-PI, NSF Arctic Natural Sciences Grant (rejected – resubmission Jan 2027)**
- **National Academies NRC Research Associate Fellowship (Awarded), 2026-2027**
- **Society of Wetland Scientists Student Research Grant, 2018**

PUBLICATIONS (popular media)

- **Spera, Alina C** “Stream Dissolved Nitrogen Cycling Responds to Human Activity across the Landscape”, LTER Network Science Stories. March 2022. [URL](#).
- **Spera, Alina C.** “Thawing Out Coastal Arctic Food Webs with Long-Term Biochemical Data”, LTER Network Science Stories. September 2021. [URL](#).
- **Spera, Alina C.** “What Microfossils Can Tell Us About Sea Level Rise”, ECO Magazine Special Issue: State of the Coast, December 2018. [URL](#).

RESEARCH EXPERIENCE

Research Assistant at the University of Texas at El Paso – August 2019-August 2024

- Led small field teams in coastal Arctic aquatic sampling, deployment of shallow marine moorings and long-term aquatic instrumentation. Managed and conducted laboratory biogeochemical operations and analysis.

Participant in BREATHE 2023 Sea Ice Field School - May 2023

- Student participant for sea ice field school aboard Kronprins Haakon ice breaking research vessel with University of Tromsø (UiT). Sea ice and oceanographic sampling and data collection.

Research Assistant at Louisiana State University – August 2017-August 2019

- Led field team in coastal wetland soil sampling and conducted laboratory biogeochemical analysis for master’s thesis to determine effectiveness of local hydrological restoration.
- Contributed to daily lab operations, organized and processed soil and water samples.

Undergraduate Thesis Project, University of Miami, Miami, FL - Spring 2016

- Project for departmental honors at the University of Miami.
- *Thesis title:* “Distribution of cephalopod paralarvae in the Eastern Caribbean (U.S. and British Virgin Islands).”

Lab Assistant in National Marine Fisheries Service, Miami FL - January 2014-May 2016

- Assisted with data collection, data entry, and support for fish and invertebrate larval research projects. Completed independent research project.

HONORS AND AWARDS

- **LSU CC&E Graduate Student Symposium Award: First Place, 2019**

PUBLISHED DATASETS

1. Beaufort Lagoon Ecosystems LTER. 2026. Daily average pCO₂, dissolved oxygen, and water quality in Elson Lagoon: 2019, 2021, 2022 ver 1. Environmental Data Initiative. <https://doi.org/10.6073/pasta/bd140df496a851ced06f8f03445f73ab>.

2. Beaufort Lagoon Ecosystems LTER, Core Program. 2026. Spot and continuous seasonal pCO₂ sensor measurements from lagoon and river sites along the Alaska Beaufort Sea coast, 2019-ongoing ver 4. Environmental Data Initiative.
<https://doi.org/10.6073/pasta/fe576c1fedeb7a369c6d58c1fa3431e6>.
3. Beaufort Lagoon Ecosystems LTER, Core Program. 2025. Free water metabolism (FWM) derived net ecosystem production (NEP) from lagoon sites along the Beaufort Sea Coast, Alaska, 2019-ongoing ver 1. Environmental Data Initiative.
<https://doi.org/10.6073/pasta/74f675276213727571c2b3f6c197fa17>.
4. Beaufort Lagoon Ecosystems LTER, A. Spera, and V. Lougheed. 2025. Daily summer riverine and groundwater pCO₂, dissolved oxygen, and ecosystem metabolism from sites along the Alaska Beaufort Sea coast, 2019-2022 ver 1. Environmental Data Initiative.
<https://doi.org/10.6073/pasta/eb56765417b7156cb178c73d963c7283>.

PRESENTATIONS

Invited

1. *“Thawing Boundaries: Coupled Physical and Biogeochemical Processes Shaping Terrestrial-Aquatic Connectivity in Permafrost Landscapes”*. Permafrost Young Researchers Network Webinar Series, May 8, 2026.
2. *“Saltwater Intrusion as an Emerging Driver of Coastal Hydrologic Change in the Arctic Coastal Plain”*. University of Texas at El Paso, Ecology and Evolutionary Biology Program Seminar, April 8, 2026.
3. *“Arctic Aquatic Connectivity Under Change: Linking Physical and Biogeochemical Processes Across the River–Ocean Interface”*. Northeastern University Marine Science Department Seminar, March 10, 2026.

Conference Presentations

1. **Coastal and Estuarine Research Federation, Richmond, VA - November 2025**
Talk title: “Tundra salinization alters subsurface hydrothermal characteristics and threatens Arctic coastal resilience”
2. **Saltwater Intrusion Meeting (SWIM), Barcelona, Spain – June 2025**
Talk title: “Feedbacks between saltwater intrusion, permafrost thaw, and soil thermal properties in an arctic coastal system”
3. **Ocean Sciences Meeting, New Orleans, LA - February 2024**
Talk title: “What is the relative contribution of nearshore CO₂ flux to estuarine carbon balance in an Arctic lagoon?”
4. **Coastal and Estuarine Research Federation, Portland, OR - November 2023**
Talk title: “Seasonal trends of CO₂ flux and metabolic status of Arctic lagoons”
5. **Joint Aquatic Sciences Meeting, Grand Rapids, MI - May 2022**
Talk title: “Relating form to physical and ecological contributions to CO₂ evasion in small streams on the Alaskan Arctic coastal plain”
6. **American Geophysical Union, New Orleans, LA - December 2021**
Poster title: “Seasonal Shifts in Carbon Flux and Cycling in an Arctic Lagoon”
7. **Alaska Marine Science Symposium, Anchorage, AK - January 2020**
Poster title: “Land-Ocean Connectivity of Inorganic Carbon Dynamics in Small Coastal Watersheds and Lagoon Systems on the North Slope of Alaska”
8. **LSU CC&E Graduate Student Symposium, Baton Rouge, LA - April 2019**
Talk title: “Decadal changes of soil physiochemical properties in a freshwater wetland after hydrologic reconnection”
9. **Soil Science Society of America Annual Meeting, San Diego, CA - January 2019**

Talk title: “Effect of Mississippi River Surface Water Diversion on Wetland Soils after 10+ Years of Operation”

10. State of the Coast Conference, New Orleans, LA - June 2018

Poster title: “Effect of hydrologic restoration on coastal Louisiana freshwater wetland soil properties and resilience”

11. International Symposium on Biogeochemistry of Wetlands, Ft Lauderdale, FL April 2018

Talk title: “Effect of hydrologic restoration on coastal Louisiana freshwater wetland soil properties and resilience”

**12. University of Miami Rosenstiel School of Marine and Atmospheric - May 2016
Science Undergraduate Poster Symposium, Miami FL**

Poster title: “Distribution of cephalopod paralarvae in the Eastern Caribbean (U.S. and British Virgin Islands).”

OUTREACH & BROADER IMPACTS

- Science Field Trip Facilitator, Insights El Paso Science Center, 2022-2024
- Invited Speaker, El Paso High School Science Career Showcase, 2022-2023
- Community Outreach Volunteer, Utqiagvik Boys and Girls Club, 2022
- Outreach Coordinator, Ukpiaqvik Iñupiat Corporation Science & Culture Fair, 2019, 2023
- Louisiana Sea Grant Ocean Commotion Outreach Event Organizer, 2017-2018

TEACHING + SCIENCE COMMUNICATION

Teaching experience:

Guest Lecturer University of Texas at El Paso, 2023-2024

- Prepared two lectures for a graduate and undergraduate level Limnology course on aquatic biogeochemistry.

Guest Lecturer Virginia Institute of Marine Science, 2023

- Delivered lecture for a graduate level Arctic Sciences course on Arctic terrestrial and aquatic greenhouse gas fluxes.

Field Station Educator at Chincoteague Bay Field Station, Wallops Island VA – 2016-2017

- Lead placed based marine ecology and oceanographic education programs for children, adults, and college level field course programming.

Assistant Naturalist with NJ Fish and Wildlife, IBSP NJ - Summer 2013, 2014

- Operate hands-on, place-based environmental education programs located in Barnegat Bay Marine Conservation Zone for students and educational groups.

Pedagogical Training:

Co-Organizer for Programming in Earth Science Classrooms Educational Workshop (PECES)

- Organized workshop aiming to equip early career scientists to effectively incorporate programming into Earth Science course instruction.

ACUE Course in Promoting Active Learning - Fall 2023

- Completion of an Association of College and University Educators course requiring the implementation of evidence-based instructional approaches.

Science Communication:

- LTER Network Student Science Writing Fellow- April 2021-August 2022
- Publication Editor and Author for Envirobites.org - May 2018-August 2020
- Contributor to LSU College of the Coast and Environment Newsletter - June 2018-May 2019

MENTORING

University of Texas at El Paso

- Primary research mentor for eight undergraduate students and one post-baccalaureate researcher
- Supervised student fieldwork in remote Arctic environments, laboratory analyses, and data interpretation.
- Mentoring outcomes:
 - Two undergraduate poster presentation awards
 - One article published with post-baccalaureate researcher as first author

Louisiana State University

- Research mentor for three undergraduate students in laboratory research, data analysis, and conference presentation preparation.

SERVICE TO THE PROFESSION

- Subcommittee Co-Chair, Attendee Experience Committee, Coastal and Estuarine Research Federation Annual Meeting, 2024-2025
- Executive Committee Student Representative, Beaufort Lagoon Ecosystems (BLE) LTER, 2020-2023
- President, Biology, Environmental and Engineering Graduate Student Group, University of Texas at El Paso, 2020-2022

PROFESSIONAL MEMBERSHIP

- Society of Wetland Scientists (SWS), American Geophysical Union (AGU), Coastal and Estuarine Research Federation (CERF)
- Alpha Pi Chapter of Alpha Phi Omega, International Co-ed Service Fraternity